

Nguyen Tran Duc Hoang

Thu Duc, Ho Chi Minh City, Vietnam • hoangdelima0905@gmail.com • 0817 002 407
github.com/hoangdelima09 • linkedin.com/in/khmtbku2407

Summary

Final-year Computer Science student at HCMUT-VNU specializing in **Computer Vision**, with an **8-week AI Engineer internship at CT Group – Gascolae** on the GASCOLAE platform: built **14 responsive service landing pages** with an embedded AI-assistant widget (Tailwind CSS, JavaScript, Alpine.js) and wrote **prompt libraries and AI agent configurations** for UAV inspection services in the energy sector. Hands-on across the full computer vision model pipeline in personal projects (**object detection and multi-object tracking**), from data preparation to deployment. Seeking an **AI Engineer (Computer Vision) Fresher** position.

Education

HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY - VNU (HCMUT-VNU) Ho Chi Minh City, Vietnam

Bachelor of Computer Science. GPA: 3.2/4.0

Expected 2027

Relevant Coursework: Machine Learning, Data Mining, Programming for AI and Data Science, Data Structures and Algorithms, Database Systems.

Experience

CT GROUP – GASCOLAE

Ho Chi Minh City, Vietnam

AI Engineer Intern

Aug 2026 – Oct 2026

- Worked on the GASCOLAE platform in a 3-member team owning 14 UAV services in the Energy, Utilities & Renewable Power sector: 9 in Substation Maintenance & Safety and 5 in Smart O&M & Analytics (solar farms).
- Built 14 responsive landing pages for these services (thermal, RGB, SF₆ gas, acoustic, UV-corona, LiDAR, hyperspectral) with Tailwind CSS, JavaScript and Alpine.js; deployed the SF₆ page on Firebase Hosting.
- Researched and delivered a complete 12-asset package for each service, about one service every 5 working days: research workbook, knowledge base, service profile, metadata catalog, SOP checklist, proposal, 13-slide sales kit, pricing, AI prompt library, landing page, AI agent configuration and competitor analysis.
- Specified the platform's AI service assistants: 150 reusable prompts, 190+ test cases and guardrails for grounding, data classification and human escalation, packaged for the dev team to implement.
- Benchmarked about 30 competitors per service (e.g. Skydio, Percepto, DJI Enterprise, Cyberhawk), kept every asset traceable to cited sources, and audited all 14 packages for consistency across files.

Projects

GASCOLAE UAV Service Landing Pages (14 websites) | [Live demo](#)

Aug 2026 – Oct 2026

HTML5, Tailwind CSS, JavaScript, Alpine.js, Lucide Icons, Firebase Hosting

- Developed 14 single-page marketing sites, one per UAV service (thermal inspection, oil-leak detection, SF₆ gas sensing, LiDAR 3D mapping, solar-farm analytics, etc.), on a shared Tailwind design system with custom colour tokens, typography and reusable card, timeline and pricing-tier components.
- Structured each page into 10+ sections (hero, customer problems, capabilities, process, service levels, deliverables, FAQ, contact) with responsive layouts for mobile, tablet and desktop, plus SEO meta tags.
- Implemented the interactive parts in vanilla JavaScript and Alpine.js: an AI-assistant chat widget with suggested questions and escalation to sales, lead-capture form and modal, FAQ accordion, content-validation checklist panel and an RGB/thermal image-layer toggle.
- Turned each service's knowledge base into public web copy under strict content rules (no unapproved prices, SLA or accuracy claims) and added source traceability to every page footer.

AI Service Assistant: Prompt Library & Agent Configuration

Aug 2026 – Oct 2026

Prompt Engineering, LLM Agent Design, Guardrails, Test-Case Design, Excel

- Wrote agent configurations for 14 service chatbots: role, welcome message, 10 prioritised knowledge sources, 8–10 intents and 8–10 behaviour rules (grounding, no hallucination, pricing, security, service boundary, escalation).
- Built a library of 150 reusable prompts in 6 categories (service explanation, use-case matching, service levels, SOP, sales objections, FAQ and gap check), each with allowed sources, expected output and restrictions.
- Designed 190+ test cases covering happy path, data gaps, pricing, restricted data, cross-service and commitment requests, plus a metadata catalog with AI search rules and Public/Internal/Restricted filters.

Smart Traffic Monitoring: Vehicle Detection, Tracking and Counting | [GitHub](#)

Jul 2026 – Sep 2026

YOLO11m, Ultralytics, PyTorch, ByteTrack, OpenCV, Streamlit

- Fine-tuned YOLO11m on BDD100K (70K training / 10K validation images, 1.28M boxes, 10 classes) on two Tesla T4 GPUs with mixed precision; selected the best of 4 checkpoints at mAP@50 0.58 / mAP@50–95 0.33 on validation, with consistent test results (0.44 mAP@50–95 on the 4 vehicle classes).
- Built a data validation pipeline (corrupt-image scan, image–label pairing, YOLO label checks) that repaired 9.2K invalid boxes.
- Implemented ByteTrack/BoT-SORT tracking with persistent IDs and a directional line-crossing counter; delivered a Streamlit app that exports annotated H.264 video and per-event CSV reports.

Facial Emotion Recognition Web Application | [GitHub](#)

Sep 2025 – Dec 2025

TensorFlow, Keras, ResNet101, OpenCV, Flask

- Fine-tuned ResNet101 on RAF-DB (7 emotions) with two-stage transfer learning, class weighting and data augmentation, reaching 70.6% test accuracy; deployed it in Flask with OpenCV face detection.

IoT Smart Home Access Control | [GitHub](#)

Jan 2026 – Jun 2026

Face++ API

- Integrated the Face++ API for face recognition into a smart-home system for access control and user verification.
- Implemented camera input processing and decision logic to connect recognition results with device control.
- Integrated IoT communication components to support real-time monitoring and automated access actions.

Credit Risk Prediction | [GitHub](#)

Mar 2026 – Apr 2026

XGBoost, Scikit-learn, SHAP, Streamlit

- Built an XGBoost default-risk model on 600K+ Lending Club loans with class weighting and threshold tuning (ROC-AUC 0.742 vs. 0.716 baseline), explained with SHAP and served in a Streamlit dashboard.

Skills

Computer Vision: Detection (YOLO11, Ultralytics), multi-object tracking (ByteTrack), ResNet, OpenCV

Machine Learning: PyTorch, Scikit-learn, XGBoost, SHAP, NumPy, Pandas, Matplotlib

Programming & Tools: Python, C++, SQL, Streamlit, Flask, REST APIs, Docker, Git

Web Development: HTML5, CSS3, Tailwind CSS, JavaScript, Alpine.js, responsive design, SEO, Firebase Hosting

Generative AI: Prompt engineering, AI agent configuration (intents, guardrails, escalation), LLM test-case design

Business & Documentation: Market and competitor research, SOPs, proposals, pricing sheets, sales kits

Language: English (TOEIC Listening & Reading 610)